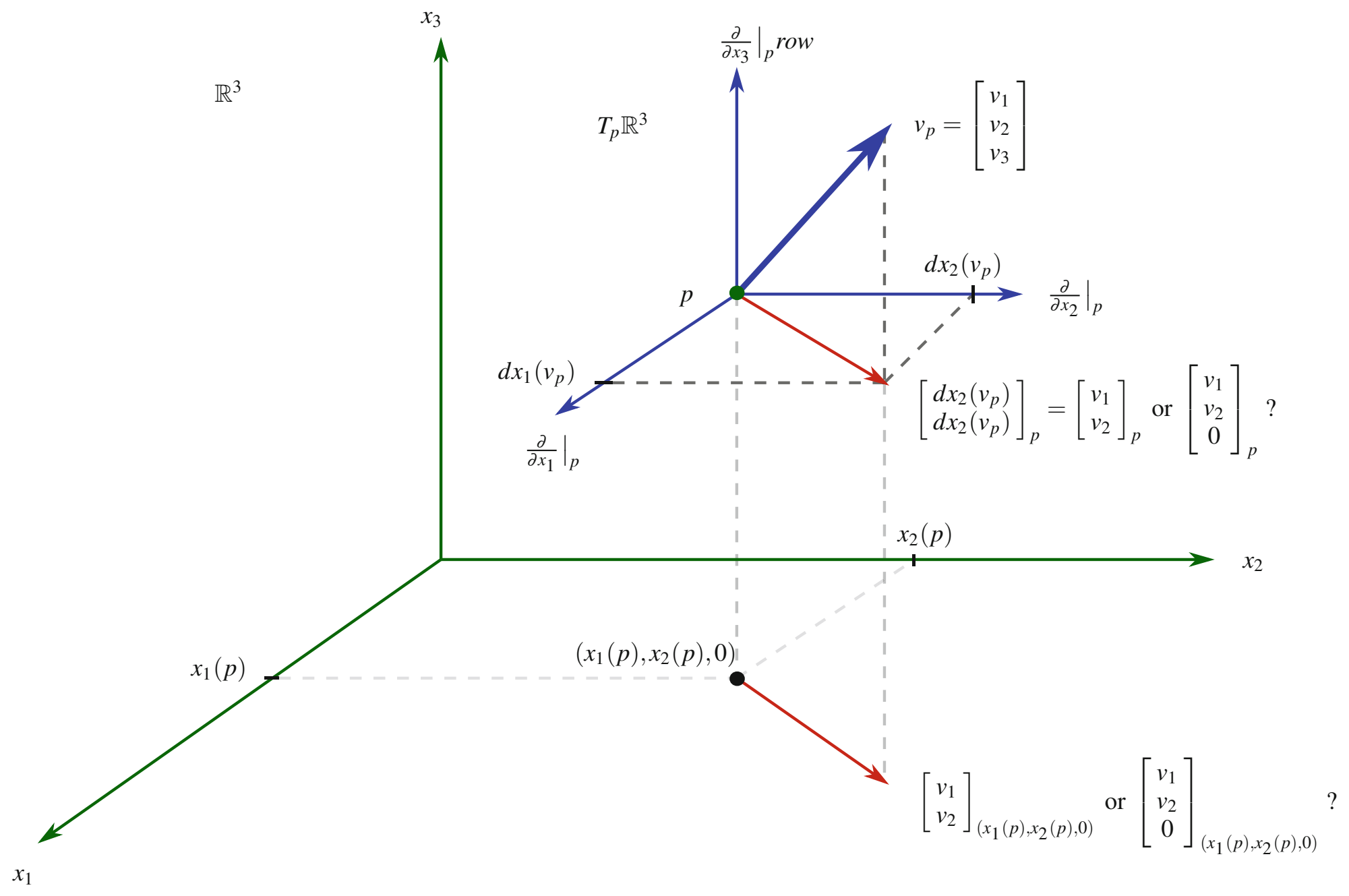

$x_3$
$\mathbb{R}^3$
$\frac{\partial}{\partial x_3}\Big|_p row$
$T_p\mathbb{R}^3$
$v_p = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$
$dx_2(v_p)$
$p$
$\frac{\partial}{\partial x_2}\Big|_p$
$dx_1(v_p)$
$\begin{bmatrix} dx_2(v_p) \\ dx_2(v_p) \end{bmatrix}_p = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}_p$ or $\begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix}_p$ ?
$\frac{\partial}{\partial x_1}\Big|_p$
$x_2(p)$
$x_2$
$(x_1(p), x_2(p), 0)$
$x_1(p)$
$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}_{(x_1(p),x_2(p),0)}$ or $\begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix}_{(x_1(p),x_2(p),0)}$ ?
$x_1$